Influence-Aware Decision Making on a HEALPix World

Esra Sekerci
Department of Information Systems
Middle East Technical University
Ankara, Turkey
esra.sekerci@metu.edu.tr

***Abstract*—This report documents a Python implementation of a game on a HEALPix sphere (NSIDE=2, 48 regions) and evaluates two agents: a baseline greedy agent and an influence-based agent. The environment is modeled as a graph induced by the HEALPix neighbor relation, with per-region resource weights in {1,3,5}. The influence-based agent computes a per-region influence score and makes a single-step decision by explicitly comparing immediate settlement value against the option value of building a road to unlock a high-influence region. We provide influence-map visualizations (Mollweide projection) with overlays for roads and settlements, and we report comparative win-rate experiments over multiple map designs and a sensitivity analysis over the decay parameter α.**

Keywords—HEALPix; healpy; influence maps; game AI; greedy baseline; option value; decision making.

I. Introduction

In many strategy games, actions can yield immediate payoff while also creating future opportunities by expanding reachability or blocking an opponent. The term project formalizes this trade-off on a simplified Catan-like game played on a small world of 48 regions corresponding to a HEALPix pixelization (NSIDE=2, nested ordering) [1]. Adjacency is defined by the HEALPix neighbor graph, allowing the world to be treated as a small undirected graph rather than a geometric surface. The objective is to implement a greedy agent and an influence-based agent, then empirically study how influence maps can support non-greedy decision making without multi-step search or reinforcement learning.

II. Methodology

A. Game Model

The game world is an undirected graph G=(V,E) with |V|=48 regions (pixels) and edges induced by HEALPix neighborhood. Each region p∈V has an immutable resource weight w(p)∈{1,3,5}. Each player starts with one randomly assigned initial region occupied by their starting settlement. A player's reachable set consists of their start region plus all regions connected via their own roads. At the start of each turn, a player collects income equal to the sum of weights of all regions where they have settlements. On their turn, the player may (i) build a settlement on a reachable unoccupied region (cost 2 coins), (ii) build a road from a reachable region to an adjacent unoccupied region (cost 1 coin), or (iii) do nothing. Roads produce no income and are constrained to exist only to make new regions reachable; redundant roads (both endpoints already reachable) are disallowed, and a road may only be built into an unowned (unoccupied) region. The game terminates when both players have no legal actions for a consecutive full round; the winner is the player with higher total coins at termination.

B. Implementation Overview

The full implementation and reproducible experiments are available as a public Colab notebook.

The implementation is organized around three components: (1) HealpixMap encapsulates weights, adjacency, and ownership state; (2) Game enforces the turn structure, validates actions, and records a full state history; and (3) agent classes implement decide_action(). Adjacency is precomputed once per game using healpy.get_all_neighbours with nested indexing, matching the project's HEALPix neighbor-graph requirement. Reachability is computed with a breadth-first search (BFS) over player-owned roads, ensuring that expansion logic remains consistent with graph connectivity. Action validation enforces (i) costs, (ii) reachability for settlements, (iii) single-ownership of edges, and (iv) the non-redundant road constraint by rejecting roads whose endpoints are already both reachable. For reproducibility, map weights are generated with explicit random seeds in batch experiments and the full per-turn state (settlements, roads, coins, last action) is logged to support post-hoc analysis and plotting.

C. Agent Design

a) Greedy Agent

The baseline agent follows the specification in: if any settlement is affordable and reachable, it immediately builds a settlement on the reachable unoccupied region with maximal w(p). Only when no settlement is available does it build a road, selecting the adjacent unoccupied region with highest weight among available road targets.

b) Influence-Based Agent

The influence-based agent follows the standard influence-map perspective used in game AI for spatial option valuation [3]. The strategic agent computes an influence score for each region p according to: I(p)=L(p)·R(p)·F(p), where L(p)=w(p), R(p) encodes reachability with a one-step decay α∈(0,1), and F(p) penalizes regions already incident to an opponent road (friction/first-mover effect). The influence map expresses both immediate value (reachable now) and one-step option value (reachable by building one road). At each turn, the agent identifies (i) the best reachable unoccupied region for settlement and (ii) the best one-road-away region for expansion, then explicitly compares these utilities to choose between settling and road building. This yields a single-step, influence-guided decision consistent with the project restriction against tree search and learning-based optimization.

Table 1 - Influence-Based Desicion Rule (per turn)

1	Compute reachable set R0 via BFS over owned roads
2	Compute influence I(p)=w(p)·R(p)·F(p) for all p.
3	best_settle = argmax_{p∈R0, unoccupied} I(p).
4	best_road = argmax_{(u→v): u∈R0, v adjacent, unoccupied, v∉R0} I(v).
5	If coins≥2 and I(best_settle) ≥ I(best_road): SETTLE(best_settle).
6	Else if coins≥1 and best_road exists: BUILD_ROAD(u,best_road).
7	Else: DO_NOTHING.

Table 1 operationalizes a one-step, influence-guided policy under the project's action constraints. At the beginning of a turn, the agent computes an influence field $I(p) = L(p) \cdot R(p) \cdot F(p)$, where $L(p) = w(p)$ is the local resource weight, $R(p) \in \{1, \alpha, 0\}$ encodes whether a pixel is currently reachable (1), becomes reachable via a single new road (α), or is out of range (0), and $F(p) \in \{1, 0.5\}$ down weights pixels adjacent to an opponent-owned road to model contention. The decision rule then compares the best feasible *settle* target among reachable empty pixels with the best feasible *road* that expands reachability (from reachable to adjacent unoccupied and not-already-reachable). This explicitly trades immediate gain (settling on a high $w(p)$) against option value (opening a high-influence neighbor next turn), while remaining computationally lightweight (BFS reachability plus a linear scan over pixels and local neighbors).

III. Visualization and Diagnostics

Influence maps are visualized using healpy's Mollweide projection utility (mollview) [2]. To support qualitative inspection, the plotted influence field uses a fixed color scale matching the theoretical range induced by w(p)∈{1,3,5} and R(p)∈{0,α,1}. Settlements are overlaid as colored circular markers (red for Player 0, blue for Player 1), and roads are overlaid as line segments between pixel centers. A wrap-around safeguard is applied during plotting: if the longitudes of endpoints differ by more than π, the line is suppressed to avoid projection artifacts (long streaks) in the flat Mollweide view while preserving spherical adjacency in the underlying game logic.

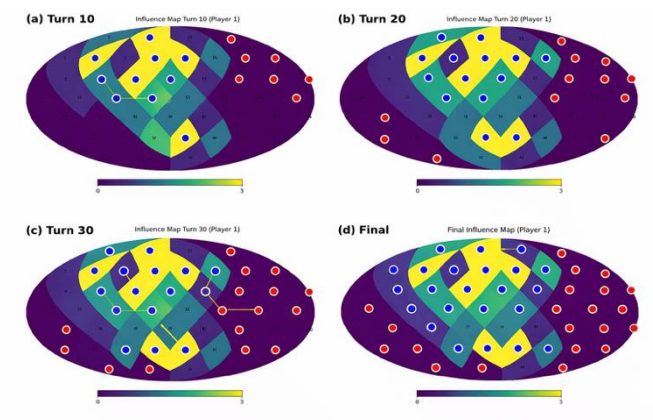

Figure 1 - Player 1 influence maps at 10, 20, 30, and final. Colors show I(p); markers/lines indicate settlements and roads (red: PO, bule: P1)

Figure 1 illustrates how the influence-based agent's "attention" evolves as reachability changes. Early in the game (Turn 10), high-intensity regions concentrate around the current reachable component and its one-road frontier, reflecting $R(p) = 1$ on owned-reachable pixels and $R(p) = \alpha$ on immediate expansion candidates. By Turns 20–30, the frontier shifts as new roads add pixels to the reachable set; influence correspondingly migrates toward newly exposed high-weight areas, while pixels near opponent roads are suppressed by the friction term $F(p)$. In the final snapshot, hotspots persist primarily where (i) weights are high and (ii) the agent's road network has maintained access, indicating that the policy effectively prioritizes securing high-value basins and their immediate expansion corridors rather than diffusing effort across low-return areas.

IV. Experimental Results

Experiments follow the three-week structure prescribed by the projec. Week 2 includes an illustrative match between a greedy agent (Player 0) and an influence-based agent (Player 1) with influence snapshots at turns 10, 20, and 30 (Figure 1). Week 3 reports comparative evaluation using batches of 50 silent games per scenario. To probe how map structure affects performance, three weight generators are used: random (uniform sampling from {1,3,5}), clustered (several high-value centers with medium-value neighborhoods), and sparse (few high-value outliers among low weights). Finally, a sensitivity sweep varies α to quantify the impact of one-step option valuation.

Table 2 - Head-to-head results over 50 simulated games for greedy and influence-based agents (including clustered and sparse variants)

Scenario (50 games)	P0 wins	P1 wins	Draws
Greedy vs Greedy	31	19	0
Greedy vs Influence	29	21	0
Influence vs Influence	26	24	0
Greedy vs Influence (Clustered)	26	23	1
Greedy vs Influence (Sparse)	29	21	0

Table 2 shows that the greedy baseline dominates overall, but the relative gap depends on map topology. On the random map, greedy outperforms influence (29 vs. 21 wins), consistent with the absence of spatial structure: when weights are i.i.d. over $\{1, 3, 5\}$, selecting the maximum-weight reachable settlement (greedy) is difficult to beat with a one-step frontier heuristic. On the clustered map, influence improves relative performance (23 wins and 1 draw vs. 26 greedy wins), suggesting that explicitly valuing the one-road frontier helps when high-value regions exhibit spatial autocorrelation (medium-weight neighborhoods around high-weight centers). On the sparse map, influence reverts to the random-like outcome (21 wins), indicating that isolated "spikes" of value are not reliably captured by a single-step reachability bonus. The greedy-vs-greedy asymmetry (31 vs. 19) also indicates a measurable player-order/initialization advantage under identical policies, which should be considered when interpreting head-to-head results.

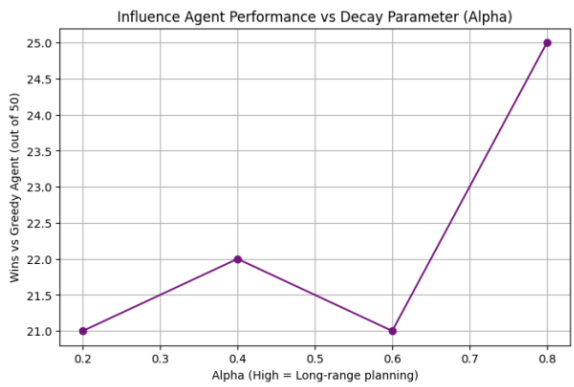

Figure 2 - Sensitivity of the influence-based agent to the decay parameter α, shown as mean terminal score (coin) advantage across 50 games

Figure 2 characterizes sensitivity to the decay parameter α, which controls how aggressively the influence agent values pixels that become reachable after building one road.

Lower αeffectively collapses the policy toward "settle-now" behavior (roads are undervalued), while higher αincreases the option value of expansion and encourages earlier road construction toward promising frontiers. The observed improvement at higher $\alpha$is consistent with the game's income mechanism (coins accrue each turn as the sum of owned settlement weights): investing in expansion sooner can increase the long-run slope of coin accumulation by enabling additional settlements on strong regions.

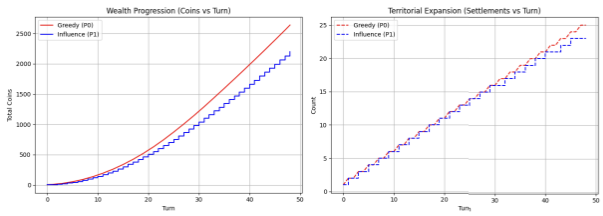

Figure 3 - Diagnostics of game progression: (a) coin accumulation versus turn and (b) settlement growth versus turn

Figure 3 provides a sanity-check linkage between mechanics and outcomes. Coin trajectories grow approximately piecewise-linearly because, at each turn, a player collects income equal to the sum of weights at all owned settlements; thus, the slope increases when a new settlement is added and remains stable between settlement events. The settlement-count curve is stepwise by construction, with plateaus corresponding to periods where the agent either (i) cannot afford a settlement, (ii) must invest in roads to unlock new reachable empty pixels, or (iii) has saturated its reachable component. Together, these diagnostics validate that the simulator's resource collection and action constraints produce the expected macro-dynamics, and they help explain why policies that accelerate access to high-weight clusters can translate into compounding coin advantages over time.

V. DISCUSSION

Across all scenarios, the greedy policy is a strong baseline because it directly optimizes the dominant short-horizon signal in this environment: immediate settlement on the maximum-weight reachable pixel, with roads used only when they lead to an immediate high-weight neighbor. The influence-based policy introduces two additional biases, frontier valuation via $\alpha$and contention modeling via friction $F(p)$, which are most beneficial when the map exhibits exploitable spatial structure. This is consistent with the clustered-map result, where "good areas" are contiguous and a one-step reachability bonus better approximates the downstream value of opening a neighborhood rather than a single pixel.

At the same time, the current influence formulation is intentionally myopic: $R(p)$ only distinguishes reachable vs. one-road-away vs. unreachable, so it cannot plan multi-step corridors to distant peaks (a key limitation on sparse maps). Moreover, because income compounds through settlement accumulation, small early misallocations (e.g., building a road when a strong reachable settlement exists, or vice versa) can have persistent effects on terminal coins and win rate. The α sweep reinforces this interpretation: higher αincreases expansion willingness, which can be advantageous when expansion unlocks sustained access to high-weight regions, but it also risks over-investing in roads when the marginal reachable settlement is already strong.

Finally, the observed asymmetry in greedy-vs-greedy indicates that player-order and random start positions can materially influence outcomes even when policies are identical. For rigorous comparative evaluation, future experiments should control for this by pairing matches with swapped player indices (policy A as P0 and P1) under matched random seeds and reporting aggregated performance. Beyond evaluation protocol, an algorithmic extension that remains aligned with the project scope is to generalize reachability from a one-step frontier to a short multi-step horizon (e.g., $R(p) = \alpha^{d(p)}$ for distance-to-reachability under valid road expansions), which would allow the influence agent to better target sparse high-value outliers while retaining the same influence-map interpretability framework.

VI. LIMITATIONS AND FUTURE WORKS

The current influence model is intentionally lightweight (one-step reachability) and thus cannot represent multi-hop planning (e.g., sequences of roads leading to distant high-value clusters). Additionally, players do not directly capture or remove opponent assets, reducing the strategic value of blocking and making local-greedy settlement selection comparatively strong. Future work could extend influence propagation to multiple hops using a decayed diffusion (α^d), incorporate explicit opponent modeling (e.g., predicting which pixels the opponent is likely to unlock next), and explore larger NSIDE values to study scaling behavior while maintaining the graph-based abstraction.

VII. CONCLUSION

A complete, reproducible simulator for influence-aware decision making on a HEALPix world was implemented in Python. The influence-based agent adheres to the project constraints by making a single-step choice grounded in an explicit influence function, and the provided visualizations and experiments demonstrate how map structure and α affect competitiveness against a greedy baseline. Empirically, the one-step influence model is most beneficial on clustered maps and at higher α, but further gains likely require multi-hop influence propagation and richer contest modeling.

REFERENCES

[1] K. M. Górski et al., "HEALPix: A Framework for High-Resolution Discretization and Fast Analysis of Data Distributed on the Sphere," The Astrophysical Journal, vol. 622, no. 2, pp. 759–771, 2005.

[2] Healpy Documentation, "healpy.visufunc.mollview," https://healpy.readthedocs.io/, accessed Jan. 2026.

[3] D. Mark, "Modular Tactical Influence Maps," in Game AI Pro 2: Collected Wisdom of Game AI Professionals, 2016.